

Linyi (Tiger) Hou

✉ linyh2@illinois.edu | github.com/TigerHou2 | Permanent Resident

EDUCATION

Ph.D. in Aerospace Engineering, University of Illinois Urbana-Champaign Jun 2021 - present
B.S. in AE, Minor in CompE, University of Illinois Urbana-Champaign Aug 2017 - May 2021

SKILLS

Technical celestial navigation, astrodynamics, Kalman filtering, modeling & simulation, data analysis
Collaboration technical/proposal writing, MS Office suite, L^AT_EX, Git, GitHub, Trello
Programming proficient: Python, C/C++, MATLAB | beginner: Simulink, Javascript, ADQL

WORK EXPERIENCE

University of Illinois Urbana-Champaign

Teaching Assistant (AE 202, AE 443, AE 483) Jun 2021 - May 2024

- ❖ Worked with 2-5 faculty and staff to teach lessons, hold office hours, and improve curriculum for courses in flight mechanics, systems engineering, and UAV controls.
- ❖ Assisted in developing code and instructions for a UAV lab based on C and Python.

Collins Aerospace

Software Engineering Intern May 2019 - Aug 2019

- ❖ Migrated an avionics data interpreter from 16-bit to 32-bit using C++ and Python.
- ❖ Automated weekly builds and integration in TortoiseSVN using Python.

RESEARCH

Astrodynamics and Planetary Exploration Group (APEX) Advisor: Dr. Siegfried Eggli

Interstellar Navigation and Star Identification Sep 2024 - present

- ❖ Developing star identification techniques for interstellar travel at relativistic speeds by combining traditional algorithms with clustering based on stellar classification.

Celestial Navigation to Supplement DSN Tracking Jul 2024 - Mar 2025

- ❖ Collaborating with four colleagues to assess feasibility of using celestial navigation to meet cruise phase uncertainty requirements in heliocentric and cislunar space.
- ❖ Designing sensor tasking/scheduling algorithms with novel considerations for covariance propagation and visibility constraints over the time domain.
- ❖ Implemented spherical harmonics for an open source orbit propagator in C++.

Lost in Space and Time Navigation using Variable Stars Nov 2022 - Mar 2025

- ❖ Developed an autonomous optical navigation technique based on light curve analysis of variable stars which does not require prior knowledge of position or time.
- ❖ Simulated star tracker observations and performed Monte Carlo analysis to assess the feasibility of reestablishing Earth pointing via the proposed method.

Aerospace Mission Analysis Laboratory

Advisor: Dr. Zachary R. Putnam

Lost in Space X-ray Pulsar Navigation (XNAV)

Jun 2021 - Dec 2023

- ❖ Developed algorithms to resolve spacecraft position and time from pulsar timing data.
- ❖ Created a photon time of arrival simulation tool in Python and validated results against the pulsar timing package [PINT](#) to within 50 μ s.
- ❖ Demonstrated the computational feasibility of (XNAV) in a lost in space scenario.

Velocity-based Initial Orbit Determination

Aug 2019 - Apr 2024

- ❖ Developed a novel initial orbit determination (IOD) technique using sequential range-rate measurements to distant stars.
- ❖ Assessed the performance of velocity IOD methods using Monte Carlo simulation.

Space Systems Optimization Laboratory

Advisor: Dr. Koki Ho

Architecture Trade Studies on ISRU Technologies for Human Space Exploration

Sep 2018 - Sep 2019

- ❖ Worked with two graduate students to discuss the development of a mixed-integer linear programming framework for space logistics optimization.
- ❖ Conducted literature review of in-situ resource utilization (ISRU) technologies to quantify input/output rates of technology options for an architecture optimization tool.

SKILLS

Programming & Typesetting	Python, C/C++, MATLAB/Simulink, Javascript, L ^A T _E X
Libraries & Frameworks	astropy scipy pybind11, Eigen OpenMP, NodeJS
Tools & Databases	Git, GitHub, Slurm, MongoDB, ADQL

HONORS AND AWARDS

Aerospace Engineering Alumni Advisory Board Fellowship	2024
Department of Aerospace Engineering, University of Illinois Urbana-Champaign	
Best Student Paper, 46th AAS GN&C Conference	2024
American Astronautical Society (AAS) Rocky Mountain Section	
Yee Fellowship	2021, 2024
Grainger College of Engineering, University of Illinois Urbana-Champaign	
David and Catherine Thompson Space Technology Scholarship	2020
American Institute of Aeronautics and Astronautics (AIAA)	
H.S. Stillwell Problem Solving Award	2020
Department of Aerospace Engineering, University of Illinois Urbana-Champaign	
Robert W. McCloy Memorial Award	2019
Department of Aerospace Engineering, University of Illinois Urbana-Champaign	
James Scholar	2018 - 2021
University of Illinois Urbana-Champaign	

JOURNAL ARTICLES

- [J.5] **L. Hou** et al. “Position and Time Determination without Prior State Knowledge via Onboard Optical Observations of Delta Scuti Variable Stars”. In: *The Journal of the Astronautical Sciences* 72.9 (2025). DOI: [10.1007/s40295-025-00485-8](https://doi.org/10.1007/s40295-025-00485-8).
- [J.4] **L. Hou** and Zachary R Putnam. “Autonomous Initial Orbit Determination using Sequential Range-Rate Measurements”. In: *The Journal of the Astronautical Sciences* 71.24 (2024). DOI: [10.1007/s40295-024-00442-x](https://doi.org/10.1007/s40295-024-00442-x).
- [J.3] **L. Hou** and Zachary R Putnam. “A Norm-Minimization Algorithm for Solving the Lost-in-Space Problem with XNAV”. In: *The Journal of the Astronautical Sciences* 71.6 (2024). DOI: [10.1007/s40295-023-00425-4](https://doi.org/10.1007/s40295-023-00425-4).
- [J.2] Hao Chen et al. “Multifidelity Space Mission Planning and Infrastructure Design Framework for Space Resource Logistics”. In: *Journal of Spacecraft and Rockets* 58.2 (2021), pp. 538–551. DOI: [10.2514/1.A34666](https://doi.org/10.2514/1.A34666).
- [J.1] Hao Chen et al. “Integrated In-Situ Resource Utilization System Design and Logistics for Mars Exploration”. In: *Acta Astronautica* 170 (2020), pp. 80–92. DOI: [10.1016/j.actaastro.2020.01.031](https://doi.org/10.1016/j.actaastro.2020.01.031).

CONFERENCE PROCEEDINGS

- [C.8] **L. Hou** and Siegfried Eggl. “Autonomous Optical Navigation using Astrometry and Photometry”. In: *AAS/AIAA Space Flight Mechanics Meeting*. AAS 25-248. Kaua’i, HI, Jan. 2025.
- [C.7] **L. Hou** et al. “Lost-in-space Position and Time Determination via Star Tracker Observations of Periodic Variable Stars”. In: *46th Rocky Mountain AAS GN&C Conference*. AAS-24-012. Breckenridge, CO, Feb. 2024.
- [C.6] Alec Auster et al. “Mars Ice Thermal Harvesting Rig & ISRU Laboratory (MITHRIL)”. In: *ASCEND 2022*. Aug. 2022, p. 4249. DOI: [10.2514/6.2022-4249](https://doi.org/10.2514/6.2022-4249).
- [C.5] **L. Hou** and Zachary R. Putnam. “A Norm-minimization Method for Solving the Cold-Start Problem with XNAV”. In: *AAS/AIAA Astrodynamics Specialist Conference*. AAS 22-560. Charlotte, NC, Aug. 2022.
- [C.4] **L. Hou** and Zachary R. Putnam. “Initial Orbit Determination from Sequential Line-of-Sight Velocity Measurements”. In: *AAS/AIAA Astrodynamics Specialist Conference*. AAS 22-561. Charlotte, NC, Aug. 2022.
- [C.3] **L. Hou**, Kevin Lohan, and Zachary R Putnam. “Comparison and Error Modeling of Velocity-Based Initial Orbit Determination Algorithms”. In: *AAS/AIAA Space Flight Mechanics Meeting*. AAS 21-280. Virtual, Feb. 2021.
- [C.2] Hao Chen et al. “Multi-Fidelity Space Mission Planning and Space Infrastructure Design Framework for Space Resource Logistics”. In: *AIAA Propulsion and Energy 2019 Forum*. 2019, p. 4134. DOI: [10.2514/6.2019-4134](https://doi.org/10.2514/6.2019-4134).
- [C.1] H Chen et al. “Integrated Analysis Framework for Space Propellant Logistics: Production, Storage, and Transportation”. In: *Lunar ISRU 2019-Developing a New Space Economy Through Lunar Resources and Their Utilization* 2152 (2019), p. 5003. URL: <https://www.hou.usra.edu/meetings/lunarisru2019/pdf/5003.pdf>.